CertBytes
(Certification Bytes)

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Some Pre-requisites

- SQL
- Python
- PySpark
- Dimensional Modeling – Fact, Dimension Tables.

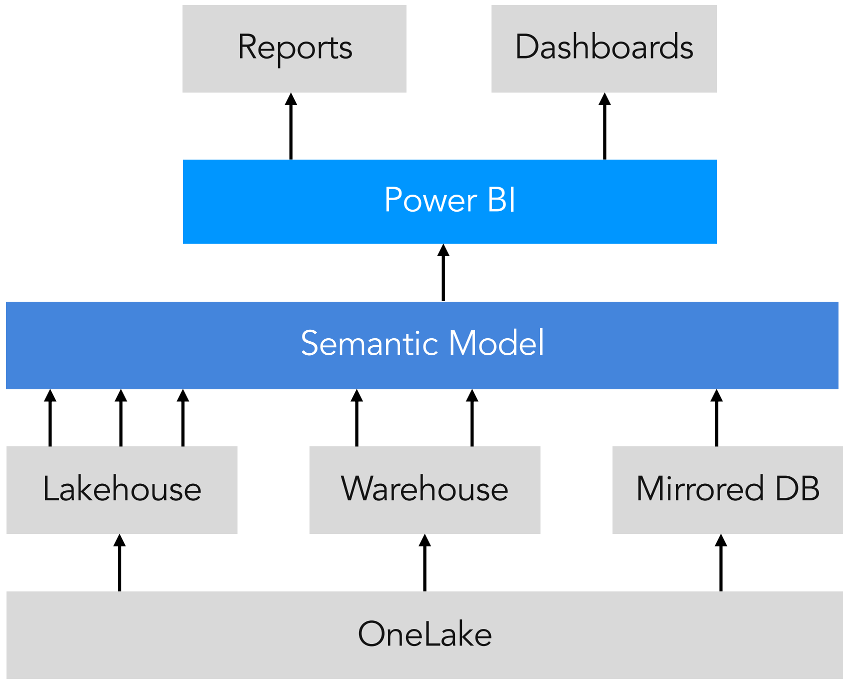
Power BI

Power BI

- Business analytics, business intelligence platform. Turn data into reports and dashboards, also known as contents.
- Queries – DAX queries. Query Engine – VertiPaq.
- Components:
  - Semantic model – Container to store schemas. Star schema, Snowflake schema, combination of multiple data sources.
  - Visualization – Charts to display data from reports & semantic models.
  - Dashboard – Multiple visuals, charts in a single place.
  - Report – Interactive visuals from a single sematic model.
  - Workspace – Contains collection of reports and dashboards.
  - Dataflow – Transformation tool in Power BI.
  - Workbook – Excel file uploaded to Power BI and can be used as data source for reports and dashboards.

Semantic Model

- Data Source for Power BI reports, dashboards etc.
- Data model used for Power BI reporting.
    - In Star Schema.
    - Warehouse.
    - Lakehouse.
    - Mirrored database (SQL Analytics Endpoint).



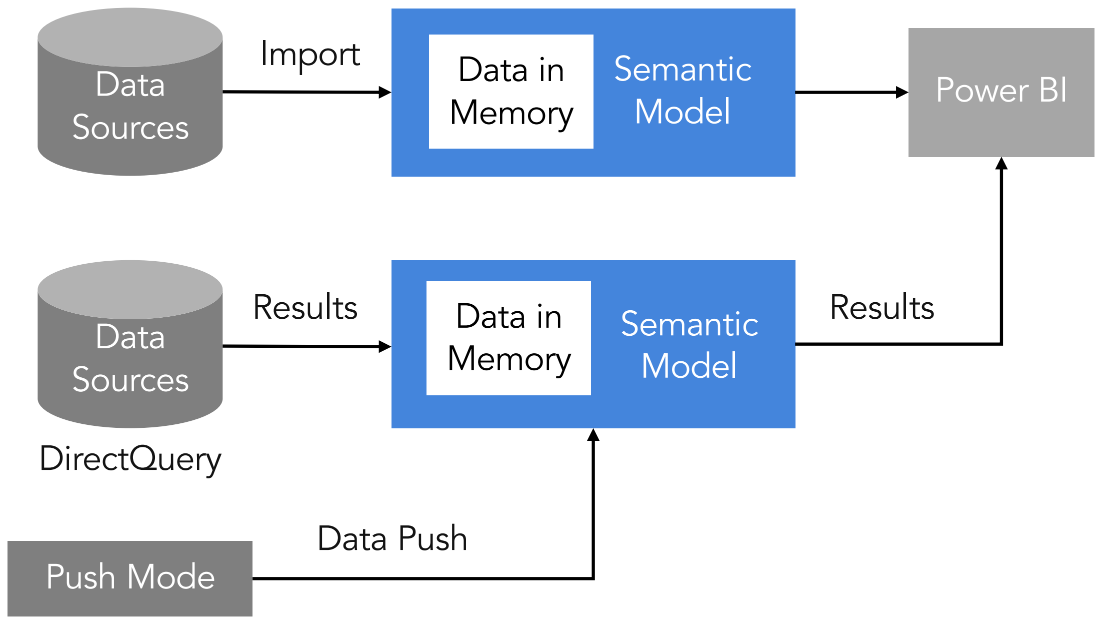
Semantic Model Permission

- Permissions:
  - Owner – User who created the semantic model.
  - Read – Allows users to read reports from semantic model.
  - Build – Allows users to build content from semantic model.
  - Write – Allows user to modify, publish, backup & restore, trigger refresh, edit semantic model settings.
  - Share – Allows users to grant access to semantic models.

	Admin	Member	Contributor	Viewer
Read	✓	✓	✓	✓
Build	✓	✓	✓	✗
Reshare	✓	✓	✗	✗
Write	✓	✓	✓	✗

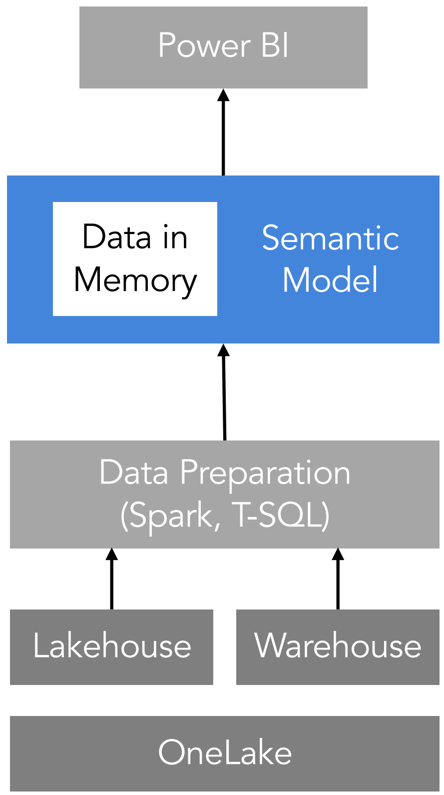
Storage Mode, Data Refresh

- Import mode.
  - Copies complete data from source to Semantic Model memory.
  - Needs semantic model refresh to fetch data changes from sources.
  - Full or incremental (partial) refresh.
- DirectQuery mode.
  - Queries are federated to sources.
  - DAX queries are converted to SQL and sent to data sources.
- Push mode.
  - No Refresh.



Storage Mode, Data Refresh

- Direct Lake mode. Storage mode for Power BI in Fabric.
    - For large tables and data volumes that cannot be imported.
    - Data loading from OneLake delta tables to semantic model memory.
    - Used when import data is challenging because of large tables.
    - Data is loaded directly into memory on-memory.



Power BI Security

- RLS (Row Level Security).
    - Restrict data for users using row filtering.
- Object Level Security (OLS)
    - Security for tables and columns.

Fabric DevOps

Lifecycle Management

- CI/CD for Fabric items.
- Git Integration (Continuous Integration).
  - Azure DevOps & GitHub.
  - For GitHub integration Fabric Premium capacity license is needed.
- Deployment Pipelines (Continuous Deployment).

Git Integration

- Using GitHub (cloud versions, NOT on-premise), Azure DevOps, GitHub Enterprise.
- Git integration is at workspace level.
- Connection and disconnection to Git Repo can only be handled by Workspace Admin.
- Adding branch to Git repo can only be done by Workspace Admin.
- After connecting any other workspace role can be used to work with GitHub.
- Authentication :
    - Azure DevOps – OAuth 2.0 and Service Principal.
    - GitHub – Personal access token.
- Can be connected to only one branch.
- Changes can be committed from Workspace to Git.
- Modifications from Git can be committed to Workspace.

Git Integration Permissions

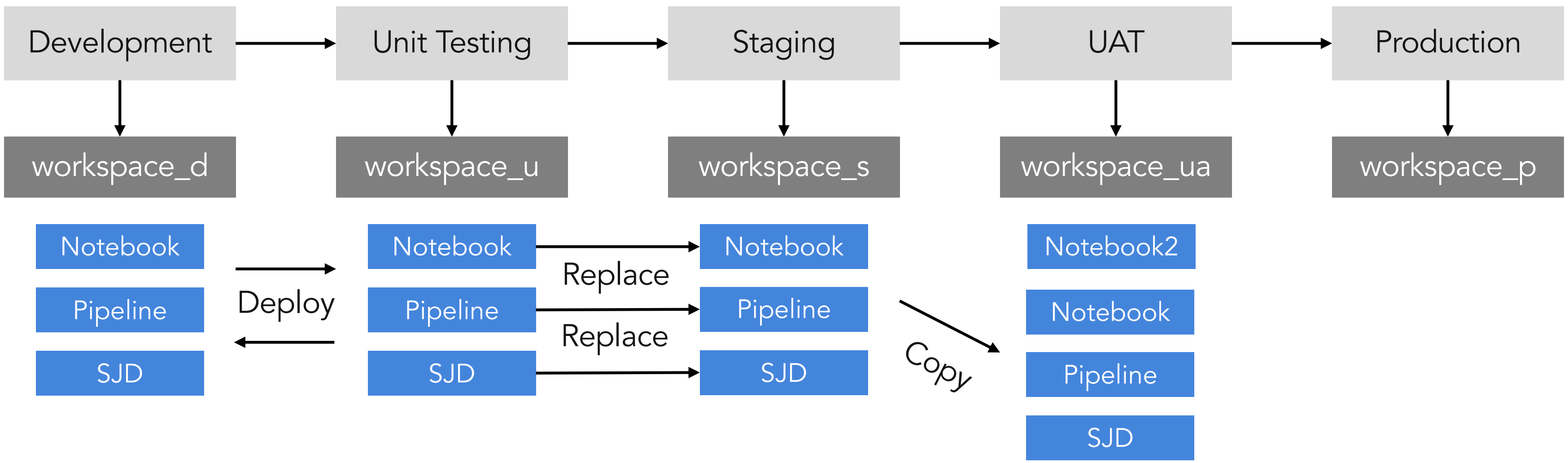
- Workspace roles.
- Git roles.

Operation	Git permissions
Connect workspace to Git repo	Contents= Access: Read
Sync workspace with Git repo	Contents= Access: Read
Disconnect workspace from Git repo	No permissions are needed
Switch branch in the workspace (or any change in connection setting)	Contents= Access: Read (in target repo/directory/branch)
View Git connection details	Contents= Access: Read or None
See workspace 'Git status'	Contents= Access: Read
Update from Git	Contents= Access: Read
Commit workspace changes to Git	Contents= Access: Read and write branch policy should allow direct commit
Create new Git branch from within Fabric	Contents= Access: Read and write
Branch out to another workspace	Content=Read and write

Operation	Workspace role
Connect workspace to Git repo	Admin
Sync workspace with Git repo	Admin
Disconnect workspace from Git repo	Admin
Switch branch in the workspace (or any change in connection setting)	Admin
View Git connection details	Admin, Member, Contributor
See workspace 'Git status'	Admin, Member, Contributor
Update from Git	All of the following roles: Contributor in the workspace (WRITE permission on all items) Owner of the item (if the tenant switch blocks updates for nonowners) BUILD on external dependencies (where applicable)
Commit workspace changes to Git	All of the following roles: Contributor in the workspace (WRITE permission on all items) Owner of the item (if the tenant switch blocks updates for nonowners) BUILD on external dependencies (where applicable)
Create new Git branch from within Fabric	Admin
Branch out to another workspace	Admin, Member, Contributor

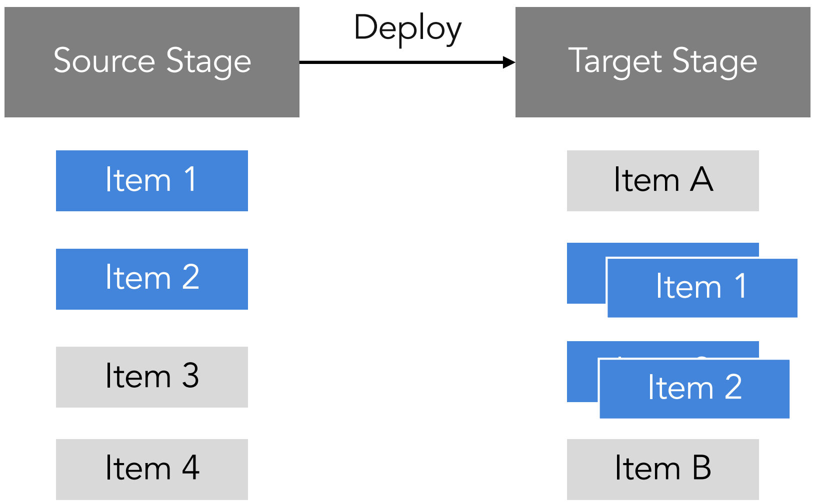
Deployment Pipeline

- Deploy across stages. Development to Production. 2 – 10 stages possible.
- Can be automated using deployment pipeline APIs.
- There can be only 1 workspace per stage.
- Clone the content of one stage to another.



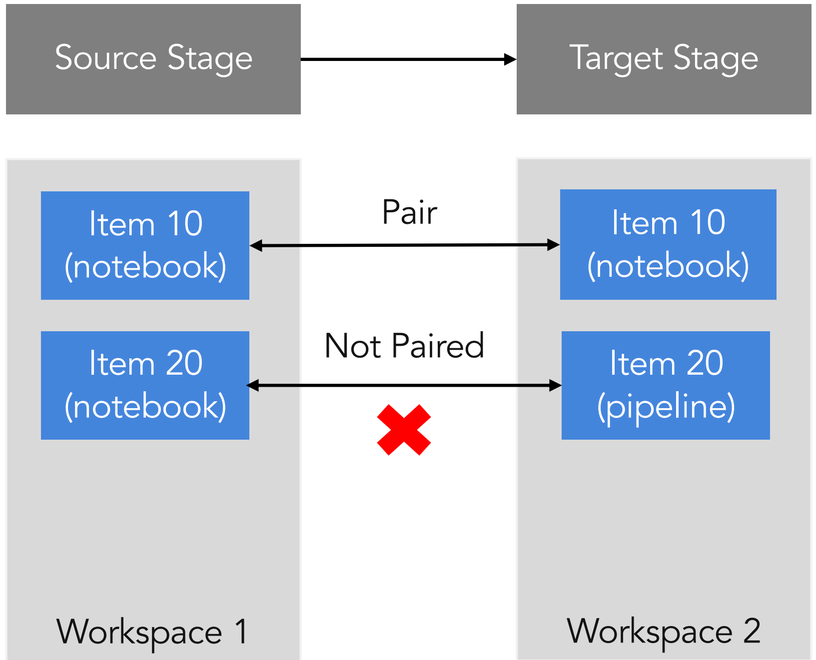
Deployment Process

- Common items on target are overwritten.
- What are copied:
  - Parameters are copied.
  - Data is not copied.
  - Metadata is copied.
  - Item permissions are not copied.
  - Workspace settings are not copied as each stage has its own workspace.
- Stages can be compared.
  - Paired items are compared automatically.
  - Green indicator – No difference.
  - Orange indicator – Items are different (modified, added or deleted).



Item Pairing

- Applies to source and target workspaces containing existing items.
- Item Pairing – Associating an item in source and target stages.
  - Item name.
  - Item type.
  - Folder location.
- Manual pairing can't be done.
- Deployment overwrites paired items.



Deployment Pipeline Permissions

- Pipeline Admin – View, edit, share, delete, add and remove workspaces from a stage, view workspaces.
- Workspace Viewer – Unassign workspace from a stage.
- Workspace Contributor – Compare stages. Deploy items (have to be contributor in both workspaces).
- Workspace Admin – Assign workspace to a stage

Fabric Workloads

Data Factory

- AWS : RDS Oracle DB, RDS SQL Server DB, S3, Redshift.
- Azure : ADLS Gen2, Cosmos DB, Azure PostgreSQL, Azure MySQL, Azure SQL Database
- GCP : Google Cloud Storage (GCS), BigQuery.
- On-premises Data Center : ODBC, JDBC, DB2, Oracle, PostgreSQL, SQL Server,
  - Using on-premises data gateway – Dataflow Gen2, Data Pipeline, Copy Job, Copy Activity.
- Azure & Power Platform : Using VNet (Virtual Network) Data Gateway.
  - Can be used with private endpoints to communicate without internet.
  - Power Apps like Dataverse, Dynamics 365 etc.
- CI/CD – Git Integration (Azure DevOps & GitHub) & Deployment Pipelines.
  - Copy Job, Copy Activity, Data Pipeline.

Data Factory

- SQL Script Activity : Execute SQL script in the data store. Fabric/Azure SQL DB, Fabric DWH, Azure PostgreSQL, Oracle, SQL Server, Snowflake, Azure SQL MI.
- Stored Proc Activity : Execute Stored Processing in Fabric/Azure SQL DB, Fabric DWH, Azure SQL MI, SQL Server DB, Synapse Analytics, RDS.
- Dataflow Gen2 Destinations: Azure SQL DB, Fabric Lakehouse Tables, Fabric Lakehouse Files, Fabric DWH, KQL Database.
- Dataflow Incremental Refresh – Need a datetime column in source that tracks the last modification.

Database & Data Warehouse

- Fabric SQL Database is offered as Operational Data Store (ODS). OLTP.
- Data from SQL Database is automatically made available in OneLake in Delta format.
- Provides SQL Analytics Endpoint.
- Provides Git Integration - Fabric to Git and Git to Fabric sync.
- Data Warehouse CI/CD integration:
  - Git – IDE (VS Code, SQL Server MS)
  - Deployment pipeline
- DWH projects can be created in VS Code using SQL database project and SQL Server (mssql).
  - SQL Database project is a grouping of SQL objects for a single database (tables, functions & stored procedures). Can contain DDL and DML statements.
  - Create new database project or from an existing warehouse.
  - After building or modifying the project can be published to warehouse.
  - SQL database projects Can be deployed from Git.

T-SQL Stored Procedure

```
CREATE PROC | PROCEDURE schema_name.procedure_name
    @parameter data_type OUT, @parameter data_type OUT
AS
BEGIN
   sql_statement;
END;
GO
```

```
CREATE PROCEDURE dbo.usp_add_kitchen @dept_id INT, @kitchen_count INT NOT NULL
WITH EXECUTE AS OWNER, SCHEMABINDING, NATIVE_COMPILATION
AS
BEGIN ATOMIC WITH (TRANSACTION ISOLATION LEVEL = SNAPSHOT)

UPDATE dbo.Departments
SET kitchen_count = ISNULL(kitchen_count, 0) + @kitchen_count
WHERE ID = @dept_id

END;
GO
```

Data Engineering

- Autotune – Autotune automatically adjusts Spark configuration to speed up workload execution and to optimize overall performance.
- Autotune uses historical execution data from the workloads to iteratively discover and apply the most effective configurations for a specific workload.
- Enable Autotune – spark.ms.autotune.enabled
  - SET spark.ms.autotune.enabled=TRUE
  - spark.conf.set('spark.ms.autotune.enabled', 'true')
- Spark settings by Autotune:
  - spark.sql.shuffle.partitions
  - spark.sql.autoBroadcastJoinThreshold
  - spark.sql.files.maxPartitionBytes
- By default spark jobs can get errored out because of capacity.
- Queueing – Spark jobs are added to queue and retried when capacity is available again. FIFO.

[TooManyRequestsForCapacity] HTTP Response code 430: This Spark job can't be run because you have hit a Spark compute or API rate limit.

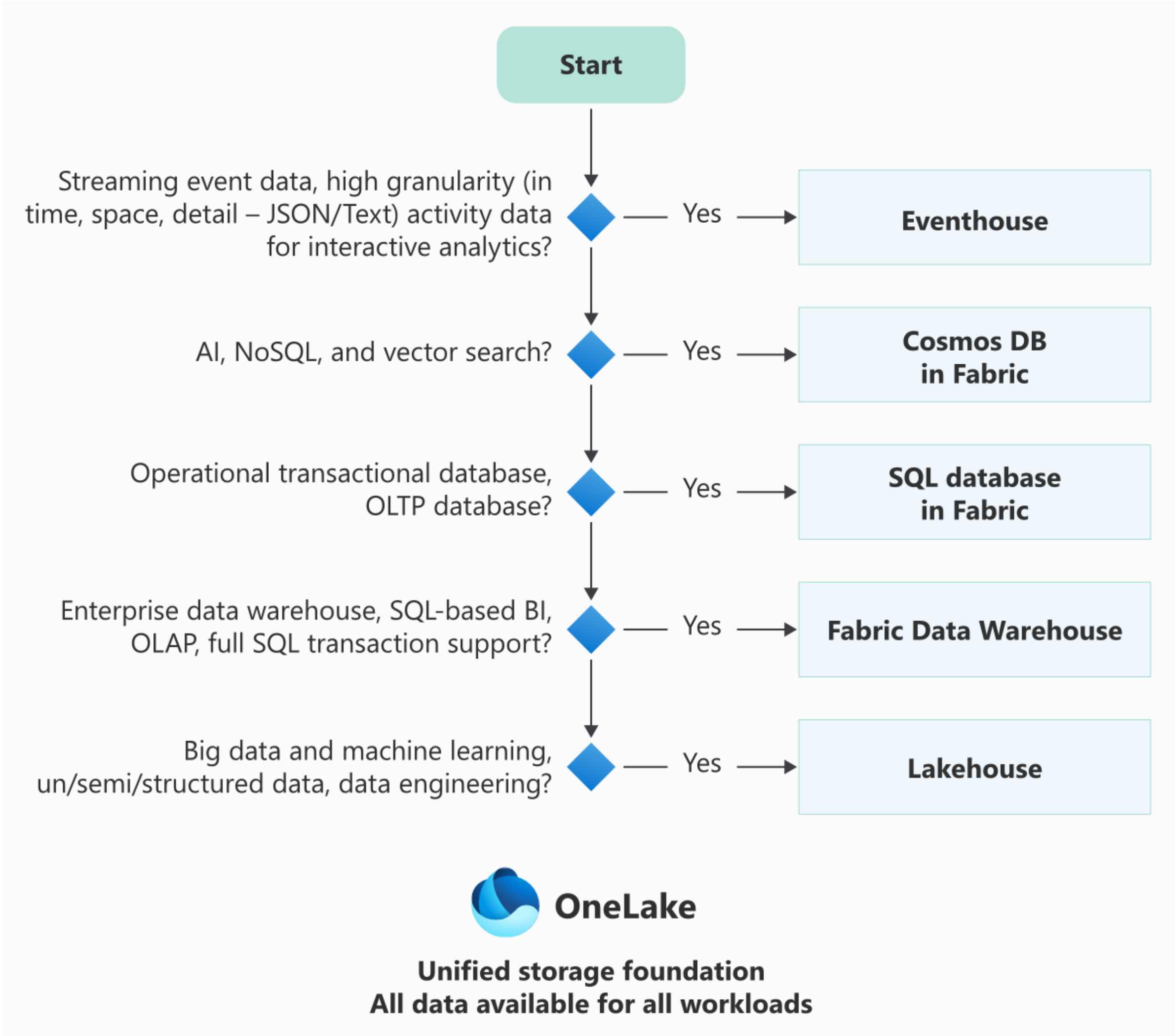
Data Engineering

- Automatic Starter Pools. Can be configured.
- Custom Pool. Node size, autoscaling, dynamic executor allocation.
- NotebookUtils Credentials:
  - notebookutils.credentials.getToken('audience Key') ; audience key – storage, pbi, keyvault, kusto.
  - notebookutils.credentials.getSecret('https://<name>.vault.azure.net/', 'secret name')
- OneLake Shortcuts & Caching:
  - Shortcuts (Lakehouse & KQL Database) point to other storage locations.
  - Retention period – 1 to 28 days.
  - File size more than 1 GB are not cached.

Real-Time Intelligence

- Eventstream data formats – JSON, CSV, Avro.
- Event processor.
- Settings – Retention period, Event throughput.
- CI/CD:
  - Using Workspace Git integration.
  - Git Integration – Commit Eventstream changes to Git and pull Git changes to Eventstream.
- Private Link and Managed Private Endpoint works with Eventstream.

Which Datastore



Which Data Integration Tool

	Data movement				Orchestration		Transformation	
	Mirroring	Copy Job	Copy Activity (Pipeline)	Eventstream	Pipeline	Apache Airflow Job	Notebooks	Dataflow Gen 2
Use Case	Data Replication	Data Ingestion & Replication	Data Ingestion	Stream Data Ingestion & Processing	Low Code Orchestration	Code-first Orchestration	Code-first Data Prep / Transform	Code-free Data Prep / Transform
Flagship Scenarios	Near real-time sync with turn-key set up. Replication.	Incremental Copy / Replication (water-mark + Native CDC), Data Lake / Storage Data Migration, Medallion Ingestion, Out-of-the-box multi-table copy.	Data Lake / Storage Data Migration, Medallion Ingestion, Incremental copy via pipeline expressions & control tables (water-mark only)	Incremental processing, event-driven and real-time AI applications.	Logical grouping of several activities together to perform a task.	Python Code-Centric Authoring	Complex Transformations	Transformation & Profiling
Source	6+ connectors	50+ connectors	50+ connectors	25+ connectors, custom endpoint (AMQP, Kafka, HTTP)	All Fabric compatible sources (depending on selected pipeline activities)	100+ connectors	100+ Spark Libraries	170+ built-in connectors + Custom SDK
Destination	Mirrored database (stored as read-only Delta table in Fabric OneLake)	40+ connectors	40+ connectors	Eventhouse, Activator, Lakehouse, custom endpoint (AMQP, Kafka, HTTP), Spark	All Fabric compatible sources (depending on selected pipeline activities)	100+ connectors	100+ Spark Libraries	7+ connectors
Type of Incoming Data	Near Real-time	Batch / Incremental Copy (water-mark based & change data capture) / Near Real-time	Batch / Bulk / Manual Watermark-based incremental copy	Real-time streaming data, change data capture feeds	All types	All types	All types	All types
Persona	Business Analyst, Database Administrator	Business Analyst, Data Integrator, Data Engineer	Data Integrator, Business Analyst, Data Engineer	Data analyst, data engineer, & integrator	Data Integrator, Business Analyst, Data Engineer	Apache Airflow Users	Data Scientist, Developer	Data Engineer, Data Integrator, Business Analyst
Skillset	None	ETL, SQL	ETL, SQL	SQL, ETL, KQL	ETL, SQL, Spark (Scala, Py, SQL, R)	Python	Spark (Scala, Py, SQL, R)	ETL, M, SQL
Coding Level	No code	No code / Low code	No code / Low code	No code / Low code	No code / Low code	Code-first	Code-first	No code / Low code
Transformation Support	None	Low	Low	Medium (stream analytics)	None	None	High	High (400+ activities)

Which Data Movement Tool

Capability	Mirroring	Copy job	Copy Activity (Pipeline)	Eventstream
Sources	Databases + 3rd party integration into Open Mirroring	All supported data sources and formats	All supported data sources and formats	25+ sources and all formats
Destinations	Tabular format in Fabric OneLake (readonly)	All supported destinations and formats	All supported destinations and formats	4+ destinations
Flexibility	Turn-key setup with fixed behavior	Simple to use + Advanced knobs	Advanced and fully customized knobs	Simple and customizable options
Custom scheduling	○	●	●	●
Table and Column management	○	●	●	●
Copy behavior: Append, Upsert, Override	○	●	●	◐
Advanced observability + auditing	○	●	●	○
Copy modes				
CDC-based continuous replication	●	●	○	●
Batch or bulk copy	○	●	●	●
Native support for Incremental copy (watermark-based)	○	●	○	○
Copy using user defined query	○	●	●	○
Use cases				
Continuous Replication for analytics and reporting	●	●	○	●
Metadata driven ELT/ETL for data warehousing	○	●	●	○
Data consolidation	○	●	●	●
Data migration / Data backup / Data sharing	○	●	●	●
Free of cost	●	○	○	○
Predictable performance	○	●	●	●

Thank You!

All the best!!